$$\hat{H}_{\mathrm{int}}(t) = \mathrm{e}^{\mathrm{i}\hat{H}_0 t}\hat{H}_{\mathrm{int}}\mathrm{e}^{-\mathrm{i}\hat{H}_0 t} = \hat{\sigma}_z \otimes \sum_i \left(g_i \hat{a}_i^\dagger \mathrm{e}^{\mathrm{i}\omega_i t} + g_i^* \hat{a}_i \mathrm{e}^{-\mathrm{i}\omega_i t} \right). \qquad (5.93)$$